

Preparation of an Ionic Compound from Aluminum Metal (Synthesis of Alum)

SONGHANG LI

(Dated: March 2023)

1. EXPERIMENTAL METHOD

Experimental method In the preparation of the experiment, 0.50 g of aluminum powder and 2.00 g KOH were placed in a 400 mL beaker. Next, 25 mL of deionized water was added into the beaker, operated in the fume hood. The mixture was stirred for about 30 minutes to be prepared in the next step. When there was no further hydrogen evaporated, the solution was filtered and collected in a 250 mL beaker. About 10 mL of 9 M H₂SO₄ solution was added to the filtered solution. The mixture was warmed by a hot plate, making the precipitate to be dissolved. The solution was tested by a litmus and shown that this was acid. The obtained solution was filtered by a Buchner funnel. The crystal was washed with the solution of 5 mL 95% ethanol and 5 mL deionized water. The final crystal was collected in the recrystallizing dish to be dried.

2. RESULT

Table I. Actual yield, theoretical yield, and percent yield for the alum synthesis

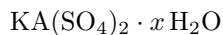
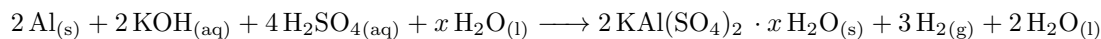
Actual Yield (<i>g</i>)	Theoretical Yield(<i>g</i>)	Percent Yield (%)
6.59	9.4	70.

Table II. Mass of alum placed into the crucible before heating and the mass of the solid in the crucible after heating

Before (<i>g</i>)	After(<i>g</i>)
0.5007	0.2499

3. DISCUSSION

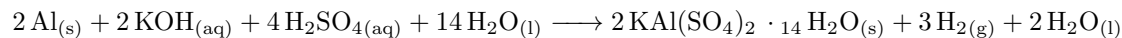
Overall reaction:



Where x is an integer. K⁺, Al³⁺, SO₄²⁻. Note: Al is limiting reactant.

The number of water was determined by calculating the mole ratio of H₂O and KAl(SO₄)₂. The weight lost after heating was the weight of water. The mole of water can be calculated by the grams of water. The weight of the solid after the heating was the weight of the KAl(SO₄)₂. Therefore, the ratio between H₂O and KAl(SO₄)₂ was $\frac{0.0139mol}{0.0010mol} = 13.9$ (rounded to 14). Therefore, KAl(SO₄)₂ · 14H₂O

The chemical equation:



The moisture from the air could be absorbed by the solid before being weighed. The actual water in the KAl(SO₄)₂ · x H₂O would be less than the measured value since the water weight in the measurement included the water from the air.